



Course: M.Sc. (IT)

Semester: 1

Prerequisite: Basic knowledge of Python programming, HTML/CSS, and fundamental web concepts is recommended.**Rationale :** develop dynamic web applications using Flask by integrating backend logic, databases, APIs, and deployment practices aligned with industry standards.**Teaching and Examination Scheme**

Teaching Scheme					Examination Scheme					Total
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
3	-	2	-	4	20	20	20	60	30	150

SEE - Semester End Examination, T - Theory, P - Practical

Course Content

W - Weightage (%) , T - Teaching hours

Sr.	Topics	W	T
1	Introduction to Flask & Basic Application Development Introduction to Web Applications and Flask framework, Installation and setup (Virtual Environment, pip), Flask application structure (App instance, routes, views), Request-Response cycle and contexts, Dynamic routing and URL variables.	18	8
2	Templates, Forms and User Interaction Jinja2 Template Engine (Rendering, variables, control structures), Static files and template inheritance, Web forms and form handling, CSRF protection, Sessions, cookies, and message flashing, Redirects and request handling	20	9
3	Database Integration & Application Structure Database concepts (SQL vs NoSQL), Flask-SQLAlchemy integration, Models, relationships, CRUD operations. Database migrations (Flask-Migrate), Application structure (Blueprints, modular design), Configuration and large application design	22	10
4	Authentication, User Management & Advanced Features User authentication (Flask-Login), Password hashing and security, User roles and permissions, User profiles and account management, Blog/posts module and pagination, Comments and social features	20	9
5	REST API, Testing & Deployment Introduction to RESTful APIs, API design and implementation in Flask, JSON handling and token authentication, Unit testing and test client, Performance optimization basics, Deployment (Cloud, Heroku, production server)	20	9
Total		100	45

Course Outcomes

At the end of this course Students Will be able to:

1	Understand Flask framework, web architecture, and application workflow
2	Develop dynamic web applications using templates, forms, and routing.
3	Design and implement database-integrated and secure web applications.
4	Build RESTful APIs, perform testing, and deploy Flask-based applications.



List of Practical

1.	Install Python, create a virtual environment, and install Flask using pip.
2.	Create a basic Flask application to display “Hello World” in a browser.
3.	Implement routing with multiple URLs and return different responses.
4.	Create dynamic routes using URL parameters and display personalized output.
5.	Create HTML templates using Jinja2 and render dynamic data in Flask.
6.	Implement template inheritance with base and child templates.
7.	Add static files (CSS/JS/images) to improve UI of the Flask app.
8.	Create a web form using Flask-WTF and handle user input.
9.	Integrate SQLite database using Flask-SQLAlchemy and create a model.
10.	Perform CRUD operations (Insert, Update, Delete, Select) using Flask app.
11.	Implement relationships between tables (one-to-many or many-to-many).
12.	Use Flask-Migrate to perform database migrations and version control.
13.	Implement user registration and login system using Flask-Login.
14.	Apply password hashing and authentication security.
15.	Create user profile page with editable information.
16.	Develop a blog module to add, edit, delete, and display posts.
17.	Create RESTful API endpoints to perform CRUD operations using JSON.
18.	Implement token-based authentication for API access.
19.	Write unit tests using Flask test client for application features.
20.	Deploy the Flask application on a cloud platform (e.g., Heroku).